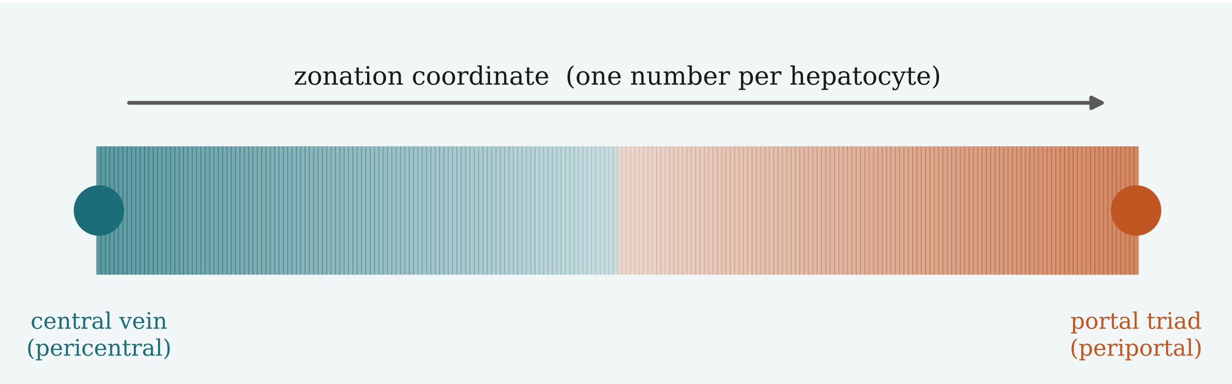
The Zonation Ruler

How it is built, normalized, and validated

A methods companion — written to answer four questions precisely:

1. The PCA maximizes the variance of WHAT, exactly?
2. How is a per-gene z-score computed — and what does “by the healthy atlas” mean?
3. What is the 8-marker validation, step by step?
4. How is the healthy quality score calculated?



Every claim here is traced to the actual code: src/steps/step4_score.py (signature scoring), step4c_learned_coords.py (PCA rulers), step4b_classifier.py (Paper-2 loader), step5_validate.py (the gate), step5b_ruler_diagnostics.py (quality metrics), and summarize_signature_battery.py (selection). Page references are to functions, not lines.

1. Orientation: what a "ruler" is

The healthy liver lobule is spatially organized along a porto-central axis. Hepatocytes near the central vein (pericentral) and near the portal triad (periportal) run opposite metabolic programs. This spatial gradient is called zonation.

Our disease data (Paper 1: 47 donors, ~69,000 hepatocyte nuclei across five MASLD stages) is single-nucleus RNA-seq — it has no spatial coordinates. A ruler is a rule that reads a cell's gene expression and returns one number: a pseudo-zonation coordinate placing that cell on the pericentral↔periportal axis. We learn the rule on a healthy reference (Paper 2: a spatial atlas that defines the zonation program), freeze it, then apply it to the disease cells and ask whether zonation degrades with disease.

> **Two families of ruler — same output**
> SIGNATURE-MEAN rulers use two published gene lists (a pericentral set and a periportal set) and average them. LEARNED rulers use PCA to discover the axis from the data with no gene list at all. Both end in exactly one coordinate per cell, and both are judged by the same healthy-quality criteria before they are trusted on disease.

The rest of this document walks the common machinery bottom-up: first how a single gene's value is normalized (Sec. 2), then the two ways genes are combined into a coordinate (Secs. 3–4), then the two healthy checks every ruler must pass — the validation gate (Sec. 5) and the quality score (Sec. 6).

2. Per-gene normalization: the z-score

A raw count matrix cannot be compared cell-to-cell or gene-to-gene directly: cells differ in sequencing depth, and genes differ by orders of magnitude in baseline expression. Two steps fix this.

Step A — depth normalization + log (per cell)

For each cell c, divide each gene's count by that cell's total counts (its library size), scale to a fixed 10,000, and take log1p. This is "log1p-CP10k":

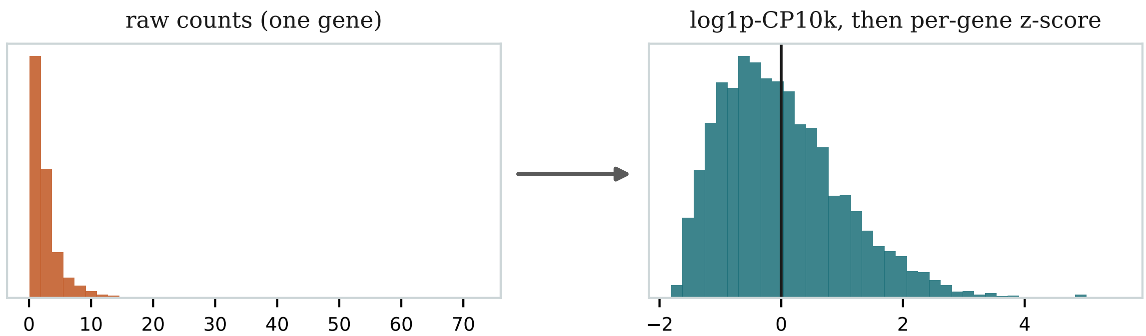
$$x_{gc} \;=\; \log\left(1 + 10^4 \cdot \frac{c_{gc}}{\sum_{g'} c_{g'c}}\right)$$

Now a value reflects relative abundance within the cell, not how deeply that cell was sequenced. (In code: np.log1p(counts / libsize * 1e4), in zrows() and _z_dense_p1().)

Step B — standardization (per gene, across cells)

For each gene g, subtract that gene's mean and divide by its standard deviation, where the mean and SD are taken over cells:

$$z_{gc} \;=\; \frac{x_{gc} - \mu_g}{\sigma_g}, \qquad \mu_g = \frac{1}{N}\sum_c x_{gc}, \quad \sigma_g^2 = \frac{1}{N}\sum_c (x_{gc} - \mu_g)^2$$



A single gene before and after. The z-score recenters every gene to mean 0, SD 1, so a highly-expressed housekeeping gene and a rare marker contribute on equal footing to the coordinate.

After this, z = +2 means “two SDs above this gene’s typical level” — a unit that is comparable across genes. Without it, a few high-abundance genes would dominate any average.

What “z-scored by the healthy atlas” means (your Q2)

The mean and SD in Step B are computed over a specific set of cells, and which set matters. For the clean label-free ruler (unsupervised_p2), μ_g and σ_g — and the PCA below — are computed entirely on the Paper-2 HEALTHY ATLAS hepatocytes (the matrix mat_norm in the atlas file, already depth-normalized by its authors, so Step A is not repeated). The resulting standardization is then applied to the Paper-1 disease cells. Because the statistics come from healthy external tissue, the ruler never ‘sees’ the disease cells while being defined — this is what keeps it leakage-free.

3. Combining genes into one coordinate: signature-mean

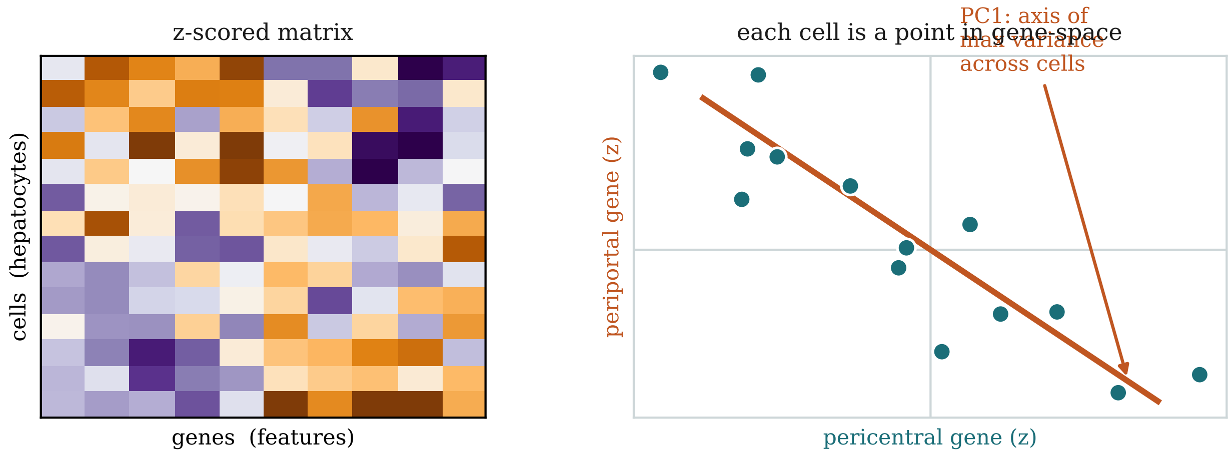
The simplest ruler takes a published pericentral gene set (PC) and periportal set (PP). For a cell, average the z-scores over each set to get two arm scores, restandardize each arm to unit variance, and subtract:

$$\bar{z}_c^{\mathrm{PC}} = \frac{1}{|\mathrm{PC}|}\sum_{g\in\mathrm{PC}} z_{gc}, \qquad \mathrm{coord}_c = \tilde{z}_c^{\mathrm{PC}} - \tilde{z}_c^{\mathrm{PP}}$$

The subtraction makes the coordinate bipolar: a cell scores high only if it is simultaneously high on the pericentral program AND low on the periportal program. Re-standardizing each arm before subtracting (the tilde) equalizes the two sides even when the gene sets are very different sizes (e.g. 26 PC vs 23 PP for the expanded set). (In code: score() in step4_score.py.)

4. The PCA ruler: variance of WHAT, exactly

The learned ruler uses no gene list. It builds the z-scored matrix Z with one row per cell and one column per feature gene (genes expressed in ≥5% of hepatocytes and present in the atlas), then runs PCA on it. Here is the crux of your question.



Left: the input is a cells×genes matrix of z-scores. Right: PCA treats each CELL as a point in gene-space and finds the straight axis along which the cloud of cells is most stretched out.

What is being maximized

PCA looks for a direction in gene-space — a vector of per-gene weights w (the loadings) — such that the per-cell score formed by that direction has the largest possible variance ACROSS CELLS:

$$s_c = \mathbf{w}^\top \mathbf{z}_c = \sum_g w_g z_{gc}, \qquad \mathrm{PC1} = \max_{\|\mathbf{w}\|=1} \mathrm{Var}_c\,(s_c)$$

Say it in words

PCA does NOT maximize the variance of a gene, and not the variance across genes. It maximizes the variance, across hepatocytes, of a single weighted-sum-of-genes score. Equivalently: of all the ways to collapse a cell's ~thousands of gene values into one number by a weighted sum, PC1 is the weighting that spreads the cells out the most. Geometrically, it is the long axis of the cell cloud.

Why does that axis turn out to be zonation? Among healthy hepatocytes — which are otherwise a fairly uniform cell type — spatial position along the porto-central axis is the single largest structured source of expression differences. So the direction of maximum cell-to-cell variance is, empirically, the porto-central axis. The genes with large positive loadings form one pole (pericentral) and those with large negative loadings the other (periportal); that the two poles are anti-correlated across cells is exactly what makes the axis a usable bipolar ruler.

We do not blindly take PC1

PCA returns several components. We compute, for each of the top components, the rank correlation between its per-cell score and a small marker axis (mean of pericentral-marker z minus mean of periportal-marker z), pick the component with the largest absolute correlation, and flip its sign so that pericentral is the high end. This guarantees the learned axis is oriented to biology, not to an arbitrary technical factor. (In code: the argmax over corrs in unsupervised()/unsupervised_p2().)

Three PCA variants — the difference is which cells the variance is taken over

- unsupervised_p2 — variance maximized over the Paper-2 healthy ATLAS cells (external to Paper 1). Leakage-free; this is the label-free co-primary ruler.
- unsupervised — variance maximized over Paper-1 HEALTHY cells. Because its healthy quality metrics are then measured on those same cells, they are in-sample

(optimistic).

- unsupervised_combined — variance maximized over the pool of Paper-1 healthy + Paper-2 atlas cells. Same in-sample caveat as above.

This is why eligibility for the frozen-primary slot is leakage-clean (axis not fit on Paper-1 cells), not publishability — see Sec. 6.

5. The 8-marker validation gate

Before any ruler is trusted, it must recover textbook zonation in HEALTHY cells. This is a positive control: if the coordinate cannot reproduce what is already known in healthy tissue, nothing it says about disease is credible. The check uses a fixed panel of eight canonical markers — the same panel for every ruler:

- Pericentral (expect to rise with the coordinate): CYP2E1, CYP1A2 (xenobiotic P450s), GLUL (glutamine synthetase — the canonical pericentral Wnt-target), PCK2.
- Periportal (expect to fall with the coordinate): ASS1 (urea cycle), ALDOB, PCK1 (gluconeogenesis), HAL (histidine catabolism).

pericentral markers expect corr(coord, gene) > 0			**periportal markers** expect corr(coord, gene) < 0		
CYP2E1	+	✓	ASS1	−	✓
CYP1A2	+	✓	ALDOB	−	✓
GLUL	+	✓	PCK1	−	✓
PCK2	+	✓	HAL	−	✓

GATE: ≥4 of 8 present AND a majority on the correct side → PASS

For each present marker we take the Spearman correlation between the coordinate and the marker's z-vector, restricted to healthy cells, and check its sign.

The rule

In healthy cells only, for each marker compute corr(coord, marker). A pericentral marker passes if the correlation is positive; a periportal marker passes if it is negative. The ruler clears the gate when at least four of the eight markers are present AND a majority of the present markers have the correct sign. The majority rule (rather than all-eight) keeps one noisy or dropout-prone marker from sinking an otherwise sound ruler. A ruler that fails is dropped before any disease analysis runs. (In code: validate() in step5_validate.py; the panel is VAL in common.py.)

> **Why this is a fair test**
> The eight markers are a fixed canonical panel, identical across all rulers, so it does not favour any particular gene list. For the label-free PCA rulers the markers are wholly independent of how the axis was built. The gate is about sign (direction), not magnitude — it asks only 'does the ruler point the right way in healthy tissue?'

6. The healthy quality score

Passing the gate makes a ruler admissible; the quality score ranks the admissible ones. It combines two healthy-cell properties, each measuring something the gate does not.

(i) PC–PP anti-correlation

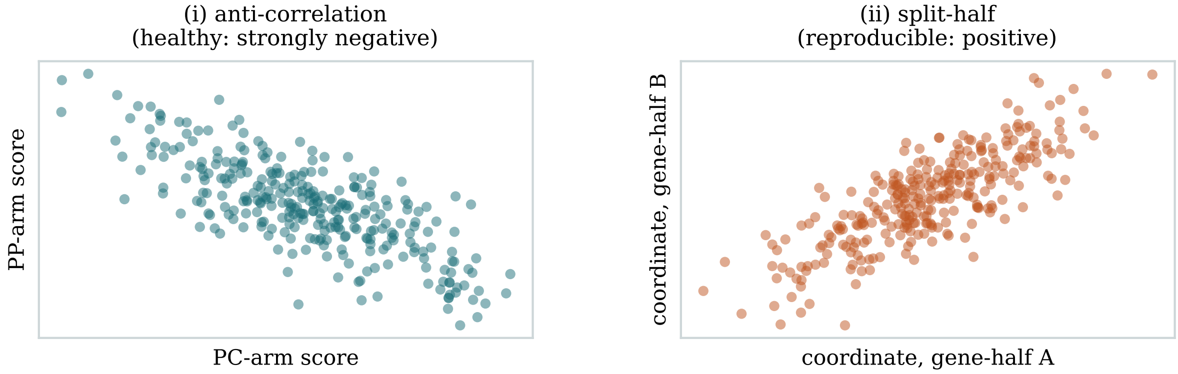
Across healthy cells, the Spearman correlation between the pericentral arm score and the periportal arm score. A genuine bipolar axis is strongly NEGATIVE: cells high on one program are low on the other. A value near zero (or positive) means the two arms are not really opposite — not a true zonation axis.

(ii) Split-half reproducibility

This asks: is the coordinate a property of the whole gene program, or an accident of a few influential genes? If the program is real, any random half of it should reconstruct almost the same coordinate as the other half. The procedure, run in healthy cells:

- Randomly split the PC genes into two disjoint halves (A and B); do the same for the PP genes.
- Build two independent coordinates: coord_A = arm(PC_A) − arm(PP_A) and coord_B from the B halves. Each is a complete little ruler made from half the genes.
- Take the Spearman correlation between coord_A and coord_B across cells.
- Repeat 20 times with fresh random splits and average; report the mean (and a 2.5–97.5% interval).

A high value (close to 1) means the two halves agree — the coordinate is reproducible and internally consistent, i.e. it is a reliable measurement, not noise. A low value means the ruler hangs on a handful of genes and should not be trusted. (In code: _coord_from_halves() and the n_splits loop in step5b_ruler_diagnostics.py.)



The two score ingredients, both measured in healthy cells. Left: a good ruler's two arms are strongly anti-correlated. Right: two coordinates built from disjoint random gene-halves land almost on top of each other (high split-half).

The combined score

Both ingredients are 'higher = better', so the score adds them, rewarding a negative anti-correlation by flipping its sign:

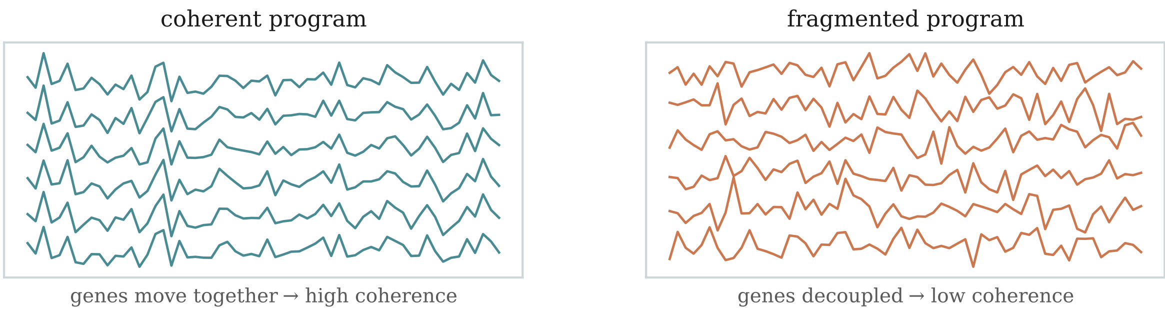
$$Q = \rho_{\text{split}} + \max(0, -\rho_{\text{PC, PP}})$$

Worked example (expanded_curated): split-half ρ = 0.670 and PC–PP anti-correlation = −0.479, so Q = 0.670 + 0.479 = 1.149. The label-free co-primary unsupervised_p2 scores Q = 0.691 + 0.471 = 1.162. (In code: the metrics are columns in ruler_diagnostics.csv from step5b; the score is healthy_splithalf_rho_mean − healthy_pc_pp_anticorr in summarize_signature_battery.py.)

What the score deliberately ignores — and the in-sample caveat (your Q1–4 tie-in)
Disease-collapse strength is NOT in the score: using the test cohort to choose the instrument would be circular. Selection uses healthy metrics only. One subtlety follows directly from Sec. 4: a PCA axis fit on Paper-1 healthy cells is then scored on those same cells, so its anti-correlation and split-half are optimistically inflated (in-sample). That is why such rulers (unsupervised, unsupervised_combined) are marked ‘control (Paper1-fit)’ and excluded from the competition, while leakage-clean rulers — published lists and Paper-2-trained axes — compete on equal footing.

(iii) A third diagnostic: within-program coherence

Coherence is reported alongside the score (and tracked into disease) but is not part of Q. It is the mean pairwise correlation among the genes of ONE program — every pericentral gene against every other pericentral gene (and likewise for periportal) — using their per-cell z-vectors within a stage. It asks a different question from the two above: not ‘are the two programs opposite?’ (anti-correlation, which is BETWEEN programs) and not ‘is the whole coordinate reproducible?’ (split-half), but ‘are the genes inside one program still moving together as a coordinated unit?’



Each line is one gene’s profile across cells. Left: a coherent program — the genes rise and fall together (high mean pairwise correlation). Right: a fragmented program — the genes have decoupled (low coherence).

Why it earns its own readout: it separates ‘the program’s genes switched off’ from ‘the program lost its internal coordination.’ Tracking coherence across disease stages helps tell a genuine loss of spatial organization from a mere change in average expression level. (In code: coherence_pc / coherence_pp via _mean_pairwise_corr() in step5b.)

7. Reading the disease results: the statistics vocabulary

The healthy checks above decide which ruler to trust. The disease tests (H1/H2/H3) then run on the 47 donors. A few recurring terms in those results deserve precise definitions.

Spearman’s ρ (rho)

ρ is a rank correlation: replace each variable’s values by their ranks (1st, 2nd, 3rd, …) and take the ordinary correlation of those ranks. It ranges from −1 to +1 and measures whether two quantities move together monotonically, without assuming the relationship is a straight line. We use it by default because single-nucleus counts and per-donor summaries are heavy-tailed and not linear, and ranks are robust to outliers. Example: ‘per-donor spread vs stage $\rho = -0.43$’ means that across the 47 donors, higher disease stage goes with smaller coordinate spread, fairly consistently. ρ is an effect size — we always report it next to a p-value, because a tiny ρ can be ‘significant’ yet biologically trivial.

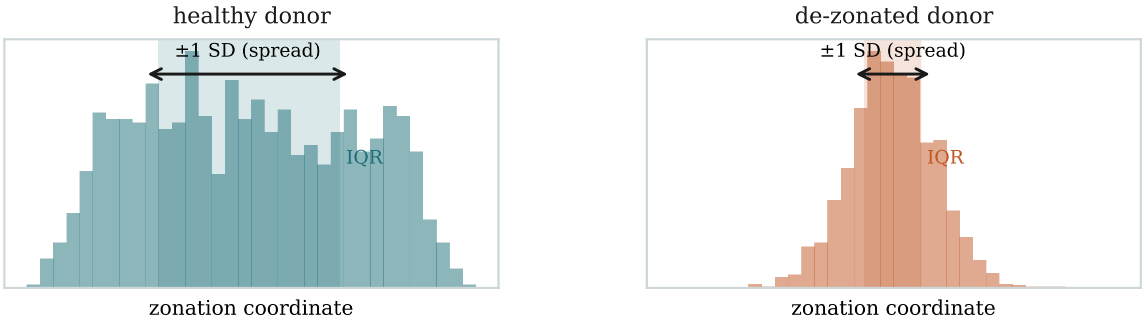
Permutation p-value (perm p)

A p-value answers: if there were truly no association, how often would chance alone produce an effect at least this large? Instead of trusting a textbook formula (which assumes normality and independence we do not have with 47 tied, non-normal donors), we compute it by SHUFFLING. We randomly permute the stage labels across donors, recompute the statistic, and repeat ~2,000 times to build a null distribution; the perm p is the fraction of shuffles whose effect is as extreme as the observed one. It assumes almost nothing about the data's shape, which is exactly what a small, messy cohort needs. (In code: perm_p in step6; jonckheere_terpstra(n_perm=2000) in common.py.)

Coordinate spread (SD) vs IQR — two ways to measure de-zonation

For one donor, collect all that donor's cells' coordinates into a distribution. A healthy donor's cells span the whole porto-central axis (a wide distribution); a de-zonated donor's cells bunch toward the middle (a narrow one). H1 measures that width two ways:

- Spread = the standard deviation of the donor's coordinates — the average distance of cells from the donor mean. Sensitive to the tails: a few extreme cells move it.
- IQR (inter-quartile range) = the 75th percentile minus the 25th percentile — the width of the middle 50% of cells. It ignores the tails, so it is robust to a handful of outlier cells or doublets.

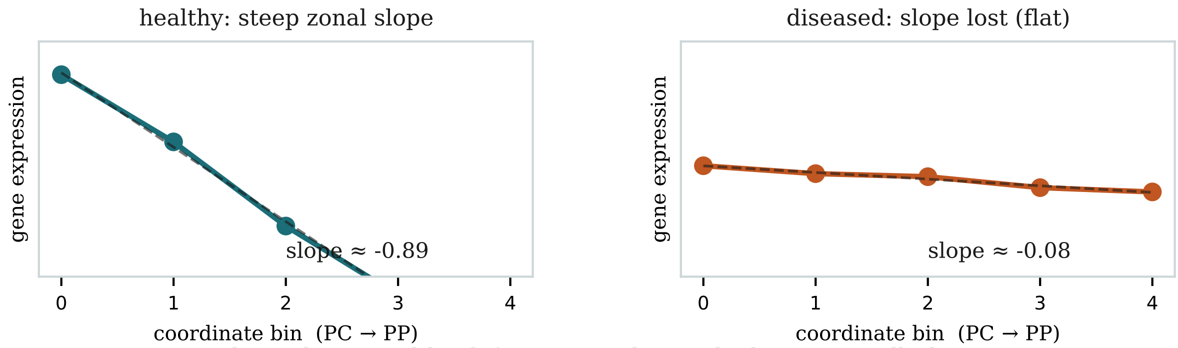


One donor's coordinate distribution. The double arrow is ±1 SD (the spread); the shaded band is the IQR. Healthy (left): wide. De-zonated (right): both shrink.

We report both precisely because they have different failure modes: if only one fell with disease, the collapse could be an artefact of that choice (e.g. SD driven by outliers). They fall together (spread $\rho \approx -0.43$, IQR $\rho \approx -0.47$ on the expanded ruler), so the collapse is real, not a metric artefact. (In code: coord_spread_std = np.nanstd, coord_iqr = p75 − p25 in step5b/step6.)

Zonal slope, and 'slope-loss, held-out' (the H2 primary test)

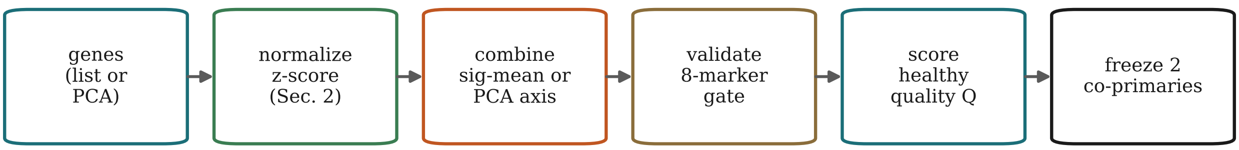
A gene's zonal slope captures how steeply it is zonated in a given donor. Bin that donor's cells into quantiles of the coordinate (five bins running pericentral→periportal), form a pseudobulk expression value per bin, and fit the slope of expression against bin index. A strongly zonated gene has a steep slope; a gene that has lost its spatial restriction has a flat one. Slope-LOSS is that slope flattening as disease advances — tested across donors after aligning each gene by its known direction so that 'more negative trend' means 'weakening' for both poles.



One gene across coordinate bins. Healthy (left): a steep slope — high pericentrally, low periportally. Diseased (right): the slope has flattened — the gene no longer tracks position. H2 asks whether this flattening trends with disease stage.

> **Why 'held-out' — avoiding a circular test**
> There is a trap: if we bin cells using a coordinate that was itself built from gene X, then test gene X's slope along that coordinate, X correlates with its own coordinate by construction — circular. The held-out design breaks this: the coordinate used for binning is built from a RANDOM HALF of the signature genes, and slope-loss is measured only on the OTHER half. No gene is ever tested against a coordinate it helped define. This is repeated over 20 random splits and averaged, so the H2 result is non-circular by construction. (In code: h2_slope_loss() with the held-out split, step7_de.py.)

8. How the pieces fit together



selection uses HEALTHY metrics only; disease H1/H2/H3 are the test

Reading left to right: choose genes (a published list, or let PCA find them); normalize each gene to a z-score (Sec. 2); combine into one coordinate per cell, either by the signature mean (Sec. 3) or the PCA axis (Sec. 4); require the coordinate to pass the 8-marker healthy gate (Sec. 5); rank the survivors by the healthy quality score (Sec. 6); freeze the best two leakage-clean rulers — one interpretable, one label-free — and only then transfer them to the disease cohort, where collapse (H1), program slope-loss (H2), and plasticity coupling (H3) are genuine out-of-sample tests.

One-line answers

Q1 PCA maximizes the variance, across hepatocytes, of a weighted-sum-of-genes score — the long axis of the cell cloud in gene-space, which is zonation.

Q2 z = (log1p-CP10k expression − gene mean) / gene SD; 'by the healthy atlas' = those mean/SD (and the PCA) come from Paper-2 healthy cells, then applied to Paper 1.

Q3 In healthy cells, check that 8 canonical markers correlate with the coordinate in the expected direction; pass if ≥4 present and a majority are correct.

Q4 Q = split-half reproducibility + the size of the (negative) PC–PP anti-correlation, both measured in healthy cells; disease is never used to select.